

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8640**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	Block:605	Seat No:	202200123
		Course Section:	1		

L_25_122_986_L_1505_41787



Name of the candidate. SUJANI GARATE

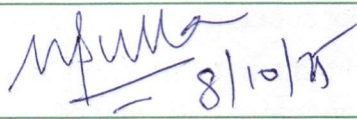
Candidate's UID number:

2	0	2	2	3	0	0	1	2	3
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Examination class room number:

5	0	8
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Invigilator's signature with date:

 8/10/25

(To be filled in by the candidate)

EXAMINATION: BE (MSE)
(For example FY MCA/FY BTECH)

SEMESTER :

PROGRAM: COMPS (B)

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: 8/10/2025 (Wednesday)

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



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	Q.3)	
	b	Gating equations of GRU: ① Update ② Reset
		Update $\Rightarrow V_i = U_i + h_{i-1}$ Reset $\Rightarrow V_i = U_i - h_{i-1}$
		How its update rule can be interpreted as concave combination of the previous hidden state and candidate easy hidden state as GRU has property to retain and propagate the previous values (outputs generated for sample) along with having the flexibility to determine how much weight is supposed to be given in each iteration to every sample. This mechanism helps to keep the relevant feature analysis intact corresponding the estimated importance (weight) of that that sample respective sample analysis. GRUs vs LSTMs GRUs : composes of 2 rules : update & reset update $\Rightarrow V_i = U_i + h_{i-1}$ Reset $\Rightarrow U_i = U_i - h_{i-1}$ LSTMs (Long short term memory) parameters : V_i, U_i, h_i, S_i, O_i LSTMs has comparatively higher parameters as it manages runs over everything in each entire set in each epoch call. for LSTMs, $V_i = U_i + S_i + h_i$ Scenarios where this might improve generalization : GRUs : Identifying Breed of Dogs. As both additive & subtractive feature matters to determine

PTG

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Scenarios:

LSTM: Handwriting recognition as is
only additive & not subtractive.

i.e. if there's confusion between 3 & 8
then it can fire ~~multiple~~ perceptions
determining probability of each.

The one with higher probability will be
forwarded as final output.

GRU: Recognising Breed of Dogs

As per per

If there's confusion between

Breed if a dog is ~~not~~ Breed A,
Breed B & Breed C and in

next layer the ear clarified that it's
definitely not B, the weightage

corresponding to features of B
will drop.

Q3
a

P10
→

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Q.3.

d.

The forget gate in LSTM self-analyzes the output after each iteration and determines the importance of each currently available feature, input data (keyword) and classifies them into ~~two~~ 2 actions i.e. to keep or to forget.

When certain input data becomes irrelevant or is itself a stopword (a, the, it) which doesn't ~~provide any contribution~~ contribute in any sort of analysis is generally classified ~~as~~ for 'to forget'.

When certain input data ~~is~~ feature is still relevant in current ~~stage~~ ^{the} n^{th} stage then the model analyzes the importance / contributing factor and determines a value of significance. This value of significance ~~deter~~ helps to either store keyword or used for further ~~for~~ mathematical computation.

It is claimed/accepted that LSTM modifies recursion and prevents vanishing gradients because it has the ability to ~~to~~ analyze and determine what to forget and what to remember corresponding to its significance factor.

The relevant data is the carried keyword till the end or ~~point to~~ ^{till} the point it seems relevant. which implies even though iterations old, the relevant data never vanishes and contributes its part to gain output determine output.

These features are ~~into~~ actively into action given that initialized with ideal/suitable parameters and mathematically customized weights and functions.

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Q.2.

b)

A linear undercomplete autoencoder trained with MSE learns a subspace equivalent to PCA as it covers all the points ~~under it~~ and determines the relevant / similar threshold values equivalent to PCA.

An over-complete autoencoder model tends to be biased as it forms rigid assumptions & strong weights to determine output.

Without regularization, the weight factor ~~can't be in same~~ will neither be in ideal scale range nor ~~will~~ have confidence significance ~~that~~ suited for respective feature.

Therefore it tends to learn the identity function where it compares the test sample's features and dataset sample having the least distance and forwards the output with this naive assumptions.

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Q.2

a)

In 2D convolution networks:

considering image is $(n \times n)$ the filter is $(f \times f)$

In each iteration a submatrix from $(n \times n)$ of size $(f \times f)$ will be chosen to undergo filter operation.

Therefore operations performed would be:

$$\frac{n}{f} + (n \% f)$$

$$\frac{n}{f} + [n \% f]$$

requiring space complexity $(n \times n) + (f \times f) + (n - f + 1)(n - f + 1)$.

⇒ High computational complexity

while in depthwise separable convolution of input size $H \times W \times C$ with K filters of size $f \times f$ the computational will increase as a new dimension has entered into consideration.

∴ operations performed would be
(operations required in 2D) $\times$ (Cw)
where Cw is new dimension

Depthwise separable convolutions are critical in modern architectures as they require more space to store and require higher high computational power as number of features & number of dimensions increase. This leads to more usage of more and advanced hardware.

Given the cost is manageable, Depthwise is preferred above rest as it is more accurate due to wide range of dimensions.

Example: In a cat detection model, the texture of fur is a significant cue apart from its shape (edges) and facial features. This fur texture is new dimension and would help to determine output accurately.



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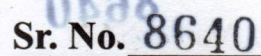
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2. b)

Overfitting occurs when model capacity is too high as + high capacity is linked to highly biased model.

It will work well on training dataset as the features and scales are rigid and redundant which might help to re-estimate accurate result.

~~It's rigidity~~ ^{rigidity} affects generalization.

[illegible]



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[illegible]

[illegible]

[illegible]